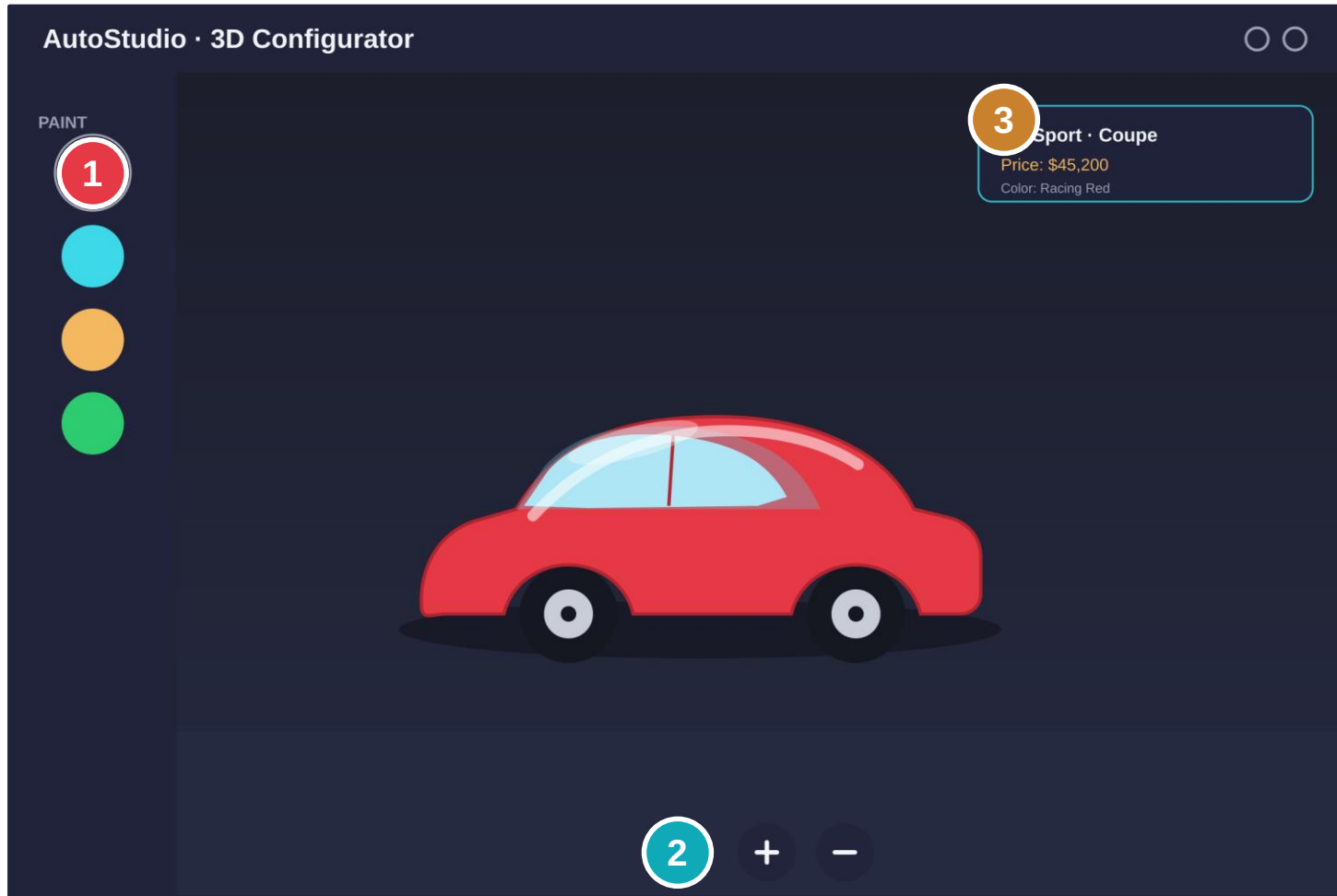


Interaction Design in a 3D Car Configurator

App: AutoStudio — an online 3D car configurator (illustrative mockup)



1

Color Swatch Picker

Clickable circles are a clear visual affordance — their shape and color signal "this is selectable," letting users change the car's paint instantly without menus.

2

Rotate / Zoom Controls

Dedicated orbit and zoom-in/out icons give users direct manipulation of the camera, so they can inspect the 3D model from any angle.

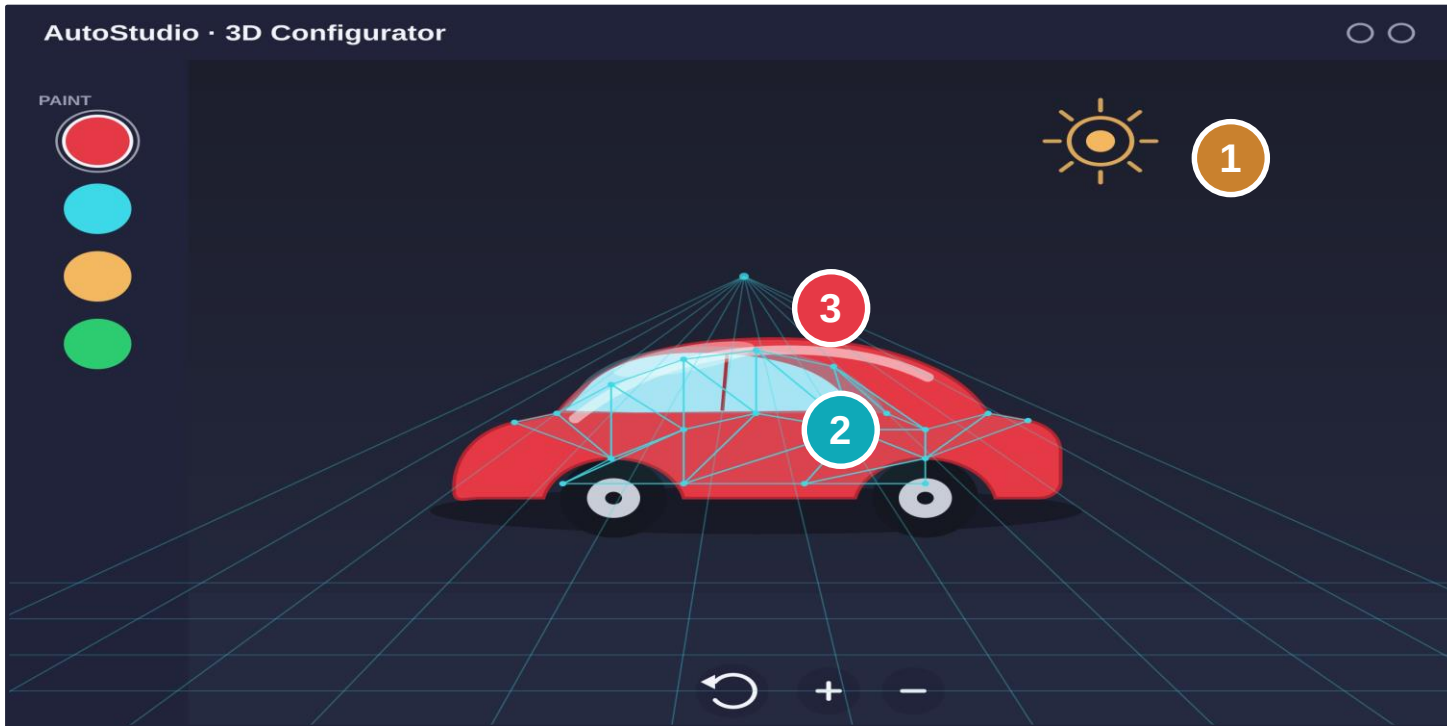
3

Dynamic Feedback Readout

The price and spec panel updates live as choices are made, giving immediate confirmation that an interaction was registered and what it changed.

Rendering the Illusion of 3D Space

Same viewport, rendering layer exposed (illustrative mockup)



1 Light Source & Shadow

A defined highlight on the hood plus a soft contact shadow beneath the car imply where light comes from, giving the flat shape volume.

2 3D Mesh Geometry

The visible triangulated wireframe is the polygon mesh underneath the painted surface — the actual 3D structure the renderer is shading.

3 Perspective Camera View

Floor lines converge toward a single vanishing point, the classic depth cue that separates a perspective camera from a flat orthographic one.

REFLECTION

1. Does CG quality help or hinder interaction? Realistic lighting and a true 3D perspective help users judge depth, proportion, and finish quality, making the car feel real. But if rendering is too heavy, slower response to rotate/zoom can make the interface feel sluggish, hindering interaction.

2. One HCI improvement: Add a text label under each color swatch (e.g., "Racing Red") on hover or tap, since color alone is not always a reliable or accessible way to identify an option.